SAHIL VAGHELA

SOFTWARE DEVELOPER

C# • C++ • Python • JavaScript • Unity • Unreal Engine • Three.js • XR

LOC Kheda, Gujarat, India TEL +91 63519 55141 MAIL vaghelasahil1104@gmail.com
IN linkedin.com/in/sahil-vaghela-ba4822250 WEB itssahilvaghela.github.io/Protfolio

// PROFESSIONAL SUMMARY

Software developer with 3+ years of professional experience building real-time 3D, VR/AR, and multiuser applications in C# (Unity) and C++ (Unreal Engine), with Python and JavaScript/Three.js for tooling and web experiences. Strengths include data-driven design, subsystem architecture, procedural geometry and DXF/JSON data pipelines, device-to-device networking, computer-vision integration, and AI-assisted development. A game-development background adds a strong focus on real-time performance, debugging, and interactive systems.

// TECHNICAL SKILLS

Languages	C#, C++, Python, JavaScript; basic C and Java
Software Engineering	Subsystem architecture, data-driven design, refactoring, debugging, API integration, DXF/JSON data processing, UI programming, Git/GitHub
Engines & Frameworks	Unity (URP, Shader Graph), Unreal Engine 5 (Blueprints, materials, lighting), Three.js, MindAR
XR & Computer Vision	OpenXR, Meta Quest 3, ARCore, Vuforia, HoloLens, MediaPipe, OpenCV
Networking	Agora Video SDK, UDP, WebSocket concepts, multiuser sessions
Tools & Design	Figma, FBX workflows, texture mapping, AI-assisted development (reviewing and debugging generated code)

// PROFESSIONAL EXPERIENCE

Unreal & Unity Game Developer | Invisible Fiction

February 2026 – Present

STACK *Unreal Engine 5, C++, Unity, C#, DXF/JSON, OpenXR, Agora Video SDK, Three.js*

- **FictionArchviz (Unreal Engine 5, C++):** developing a multiuser architectural-visualization application for desktop and Meta Quest.
- Built a floor-plan processing pipeline that extracts walls, windows, and doors from floor-plan images into separate DXF layers.
- Implemented C++ procedural geometry that reads DXF/JSON building data and creates roof meshes, including polygon-gap handling and tessellation.
- Extended the server-manager subsystem with color-coded on-screen debug output and documented the multiuser session flow.
- Developed a VR see-through/X-ray overlay material with consistent outline thickness and worked on interior-lighting balance using skylight and lighting channels.
- **MimadimHomeStudio / Mimadim (Unreal Engine):** implemented furniture data assets and a master data asset mapping furniture types to available furniture assets; added selection and replacement logic with transform preservation.
- Refactored furniture customization responsibilities toward a GameInstance subsystem to centralize selection, replacement, and future material-customization logic.
- **Fiction Mech (Unity, OpenXR, Meta Quest 3):** developed a 3D visualization experience as sole developer, including multiplayer and two-way control/communication between Quest 3 and Android using Agora Video SDK.
- **Three.js:** developed browser-based 3D game prototypes using AI-assisted coding workflows, reviewing, debugging, testing, and refining generated code.

Junior Unity Game Developer | Invisible Fiction
May 2023 – February 2026

STACK *Unity, C#, MediaPipe, OpenXR, ARCore, Vuforia*

- Developed and updated in-house Unity/C# applications, including Turbo Trail, Cupping Muffins, and Catch the Object; adapted content into themed variants and improved UI.
- Integrated MediaPipe body tracking into Alien Runner and Football Cup, streaming movement data into Unity to drive player actions.
- Worked on a QR-triggered AR model-visualization application for HoloLens using ARCore/Vuforia-related tooling.
- Contributed to Fiction Mech, a 3D model visualization application for Meta Quest 3, including OpenXR and cross-device communication work.

Artificial Intelligence Intern | Invisible Fiction
February 2023 – May 2023

STACK *Python, MediaPipe, Unity, C#*

- Prototyped human-movement recognition in Python using MediaPipe and sent tracking data to a Unity game to control the player.
- Supported day-to-day development activities and collaborated with cross-functional teammates.

// SELECTED PROJECTS

FictionArchviz `UNREAL ENGINE 5 · C++ · DXF/JSON · META QUEST`

Multiuser architectural visualization for desktop and VR, including floor-plan-to-geometry tooling, generated roof meshes, an X-ray material, lighting setup, and server-session debugging.

MimadimHomeStudio Furniture Customization `UNREAL · C++ · BLUEPRINTS · DATA ASSETS · SUBSYSTEMS`

Created furniture data assets and a master mapping from furniture type to furniture assets. Implemented model selection/replacement while preserving transforms, with architecture being organized into a dedicated customization subsystem.

Fiction Mech `UNITY · C# · OPENXR · QUEST 3 · AGORA VIDEO SDK`

3D model visualization with multiplayer and two-way interaction/control between Meta Quest 3 and Android; developed as sole developer.

WebAR Coloring Book `THREE.JS · MINDAR · GPU TEXTURE WORKFLOW`

Tracks a physical drawing sheet through a webcam and maps its live colors onto an animated 3D model. Implemented a 2048 × 2048 texture-unwarping workflow designed for real-time rendering.

Motion-Controlled Games `UNITY · C# · MEDIAPIPE`

Alien Runner and Football Cup use body/leg movement tracking to control gameplay, with movement data connected to Unity.

Unity Game Portfolio `UNITY · C#`

Turbo Trail, Cupping Muffins, Catch the Object, and related themed variants, including gameplay implementation, adaptation, and UI updates.

// EDUCATION

B.E. Computer Engineering | Ipcowala Institute of Engineering and Technology
2023 • CGPA: 9.11

HSC 2019 | 69.9% **SSC** 2017 | 84.6%

Languages: Gujarati, Hindi, English